

Autonomous Sample Retrieval Mission

Program a rover to navigate to a sample, collect it, and return to a defined home zone.

Advanced**3 class periods****Aerospace Mission**

Engineering Context

A planetary science rover has detected a high-priority sample. Communication delay prevents direct teleoperation, so the rover must complete a short autonomous sequence.

What You Will Practice

- Integrate motion, sensing, and manipulation
- Plan a mission sequence
- Use checkpoints and fault stops
- Validate a multi-step autonomous system

What You Need

- Rover platform
- Approved sensors
- Simple gripper or collection device
- Sample object
- Marked mission course
- Programming environment
- Engineering notebook

Student Instructions

- 1 Map the mission course and identify navigation checkpoints.
- 2 Create a flowchart for travel, detection, collection, return, and safe stop.
- 3 Program and test each stage independently.
- 4 Add a timeout or fault response for each critical stage.
- 5 Integrate the full sequence at low speed.
- 6 Run three official autonomous missions without code changes between trials.
- 7 Record mission time, success at each checkpoint, and failure mode.
- 8 Revise the system based on evidence.

Engineering Requirements

- No manual control after mission start
- Collect and retain the sample
- Return to the home zone
- Include at least two sensor-based decisions
- Use fault handling for missed sample or blocked path
- Complete two of three official missions

Constraints & Safety

- Maintain an emergency stop
- Use a clear enclosed course
- No high-speed impacts
- Test mechanisms separately before full integration

What You Submit

- Mission map
- Flowchart or state diagram
- Commented code
- Checkpoint test evidence
- Official mission log
- Post-mission review

Reflection Questions

- 1 Which subsystem caused the most mission risk?
- 2 How did you isolate faults during development?
- 3 Which checkpoint should be verified twice?
- 4 What would change with a communication delay?

Engineering Tip Integrate one tested subsystem at a time rather than debugging navigation, sensing, and collection simultaneously.	Common Mistake A full program that fails gives little diagnostic information if each mission stage was never tested separately.
Stretch Goal Add an alternate return route when the primary path is blocked.	Career Connection Robotics systems engineers integrate mobility, perception, manipulation, and fault management for autonomous exploration.